

Straight Line Related Problems

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Problem:

Find the equation of the ~~plane~~ ^{line} through the point $(2, 1, 3)$ and perpendicular to the plane $3x + 4y - z = 7$.

Sol:

We know the eqⁿ the line passing through the point (x_1, y_1, z_1) and having dir^s (a, b, c) is given by

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

Hence, the eqⁿ of the SL that passes through the point $(2, 1, 3)$ is given by

$$\frac{x-2}{a} = \frac{y-1}{b} = \frac{z-3}{c} \quad \text{--- ①}$$

Since line ① is perpendicular to the given plane $3x + 4y - z = 7$, so the line is parallel to the normal of that plane.

Therefore,

$$\frac{a}{3} = \frac{b}{4} = \frac{c}{-1} = k(\text{say})$$

$$\Rightarrow a = 3k, \quad b = 4k, \quad c = -k$$

Now, substituting the values of a, b, c into ① we obtain

$$\frac{x-2}{3k} = \frac{y-1}{4k} = \frac{z-3}{-k}$$

$$\Rightarrow \frac{x-2}{3} = \frac{y-1}{4} = \frac{z-3}{-1}$$

which is the required eqnⁿ of the ~~plane~~^{line}.

Problem:

Find the equation of the line through the point (1, 2, 3) and is parallel to the two planes

$$x - y + 2z - 5 = 0 \text{ \& } 3x + y + z - 6 = 0.$$

Solⁿ:

We know that the eqnⁿ of the SL passing through the point (x_1, y_1, z_1) and having dir^s (a, b, c) is given by

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} \quad \text{--- (1)}$$

Given, the eqn^s of the planes are

$$x - y + 2z - 5 = 0 \quad \text{--- (2)}$$

$$\text{\& } 3x + y + z - 6 = 0 \quad \text{--- (3)}$$

Since line ① is parallel to the planes ② & ③, so line ① is perpendicular to the normals of the planes ② & ③, respectively.

Hence,

$$a - b + 2c = 0 \quad \text{--- ④}$$

$$\& \quad 3a + b + c = 0 \quad \text{--- ⑤}$$

Solving ④ & ⑤ using cross-multiplication we get

$$\frac{a}{-1-2} = \frac{b}{6-1} = \frac{c}{1+3} = k \text{ (say)}$$

$$\Rightarrow \frac{a}{-3} = \frac{b}{5} = \frac{c}{4} = k$$

$$\therefore a = -3k, \quad b = 5k, \quad c = 4k$$

Substituting the values of a, b, c into ① we get

$$\frac{x-1}{-3k} = \frac{y-2}{5k} = \frac{z-3}{4k}$$

$$\Rightarrow \frac{x-1}{-3} = \frac{y-2}{5} = \frac{z-3}{4}$$

which is the required eqnⁿ of the SL.

Problem:

Find the eqnⁿ of the plane through the straight line $3x - 4y + 5z - 10 = 0 = 2x + 2y - 3z - 4$ and parallel to $x = 2y = 3z$.

Solⁿ:

Given,

$$3x - 4y + 5z - 10 = 0 = 2x + 2y - 3z - 4 \quad \text{--- (1)}$$

$$\& \quad x = 2y = 3z$$

$$\Rightarrow \frac{x}{6} = \frac{y}{3} = \frac{z}{2} \quad \text{--- (2)}$$

Let the eqnⁿ of the plane through the line (1) is given by

$$3x - 4y + 5z - 10 + k(2x + 2y - 3z - 4) = 0 \quad \text{--- (3a)}$$

$$\Rightarrow (3 + 2k)x + (2k - 4)y + (5 - 3k)z - 10 - 4k = 0 \quad \text{--- (3b)}$$

The plane (3b) is parallel to the SL (2), hence the SL is perpendicular to the normal of the plane.

$$6(3 + 2k) + 3(2k - 4) + 2(5 - 3k) = 0$$

$$\Rightarrow 18 + 12k + 6k - 12 + 10 - 6k = 0$$

$$\Rightarrow 12K + 16 = 0$$

$$\Rightarrow K = -\frac{4}{3}$$

Substituting the value of K into (3a) to have

$$3x - 4y + 5z - 10 - \frac{4}{3}(2x + 2y - 3z - 4) = 0$$

$$\Rightarrow 9x - 12y + 15z - 30 - 8x - 8y - 12z + 16 = 0$$

$$\Rightarrow x - 20y + 27z - 14 = 0$$

Ans

Problem:

Determine the eqn of the straight line through the origin in the plane $x - y - z = 0$ and perpendicular to the straight line $\frac{x}{2} = \frac{y}{1} = \frac{z}{-3}$.

Solⁿ:

We know that the eqn of the SL passing through the origin and having dir^s (a, b, c) is given by

$$\frac{x}{a} = \frac{y}{b} = \frac{z}{c} \quad \text{--- (1)}$$

Since eqn (1) lies on the plane $x - y - z = 0$ and is perpendicular to the SL $\frac{x}{2} = \frac{y}{1} = \frac{z}{-3}$, so

we have

$$a - b - c = 0 \text{ ————— } \textcircled{2}$$

$$\& \quad 2a + b - 3c = 0 \text{ ————— } \textcircled{3}$$

Solving $\textcircled{2}$ & $\textcircled{3}$ using cross-multiplication we get

$$\frac{a}{3+1} = \frac{b}{-2+3} = \frac{c}{1+2} = k \text{ (say)}$$

$$\Rightarrow \frac{a}{4} = \frac{b}{1} = \frac{c}{3} = k$$

$$\therefore a = 4k, b = k, c = 3k$$

Substituting the values of a, b, c into $\textcircled{1}$ we get

$$\frac{x}{4k} = \frac{y}{k} = \frac{z}{3k}$$

$$\Rightarrow \frac{x}{4} = \frac{y}{1} = \frac{z}{3}$$

which is the required eqn of the SL.

—————

$\textcircled{1}$ —

Problem : (Duplicate)

Find the eqⁿ of the plane through the line $3x - 4y + 5z - 10 = 0 = 2x + 2y - 3z - 4$ and is parallel to the line $x = 2y = 3z$.

Solⁿ :

Given,

$$3x - 4y + 5z - 10 = 0 = 2x + 2y - 3z - 4 \quad \text{--- (1)}$$

$$\& \quad x = 2y = 3z$$

$$\Rightarrow \frac{x}{6} = \frac{y}{3} = \frac{z}{2} \quad \text{--- (2)}$$

Let the eqⁿ of any plane through the line (1) is given by

$$3x - 4y + 5z - 10 + \lambda(2x + 2y - 3z - 4) = 0$$

$$\Rightarrow (3 + 2\lambda)x + (2\lambda - 4)y + (5 - 3\lambda)z + (-10 - 4\lambda) = 0 \quad \text{--- (3)}$$

If the plane (3) is parallel to the line (2), then

$$6(3 + 2\lambda) + 3(2\lambda - 4) + 2(5 - 3\lambda) = 0$$

$$\Rightarrow 18 + 12\lambda + 6\lambda - 12 + 10 - 6\lambda = 0$$

$$\Rightarrow 12\lambda = -16 \quad \Rightarrow \lambda = -\frac{4}{3}$$

Substituting the value of k into (3) we get

$$3x - 4y + 5z - 10 - \frac{4}{3}(2x + 2y - 3z - 4) = 0$$

$$\Rightarrow x - 2y + 2z - 14 = 0$$

which is the required eqn of the plane.

Problem:

Find the magnitude and the eqn of the S.D between the lines $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$ and

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}. \text{ Also find the coordinates}$$

of the points where the S.D meets the given lines.

Soln:

Given the eqns of SLs are

$$\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1} \quad \text{--- (1)}$$

$$\& \quad \frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1} \quad \text{--- (2)}$$

From ① we can write

$$\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1} = \lambda_1 \text{ (say)}$$

$$\therefore x = \lambda_1 + 3$$

$$y = -2\lambda_1 + 5$$

$$z = \lambda_1 + 7$$

Any point on line ① is given by

$$P(\lambda_1 + 3, -2\lambda_1 + 5, \lambda_1 + 7)$$

Similarly, from ② we get

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1} = \lambda_2 \text{ (say)}$$

$$\therefore x = 7\lambda_2 - 1$$

$$y = -6\lambda_2 - 1$$

$$z = \lambda_2 - 1$$

Any point on line ② is given by

$$Q(7\lambda_2 - 1, -6\lambda_2 - 1, \lambda_2 - 1)$$

Let the S.D line intersects the line ① at the point $P(\lambda_1 + 3, -2\lambda_1 + 5, \lambda_1 + 7)$ and line ②

at the point $S(7\lambda_2-1, -6\lambda_2-1, \lambda_2-1)$.

Therefore, the dir^s of the S.D line are given

$$\text{by } (a, b, c) \equiv (\lambda_1 + 3 - 7\lambda_2 + 1, -2\lambda_1 + 5 + 6\lambda_2 + 1, \lambda_1 + 7 - \lambda_2 + 1)$$

$$\Rightarrow (a, b, c) \equiv (\lambda_1 - 7\lambda_2 + 4, -2\lambda_1 + 6\lambda_2 + 6, \lambda_1 - \lambda_2 + 8)$$

Since, the S.D. line is perpendicular to the lines ① & ②, simultaneously

$$\therefore (\lambda_1 - 7\lambda_2 + 4) \cdot 1 + (-2\lambda_1 + 6\lambda_2 + 6)(-2) + (\lambda_1 - \lambda_2 + 8)(1) = 0$$

$$\Rightarrow \lambda_1 - 7\lambda_2 + 4 + 4\lambda_1 - 12\lambda_2 - 12 + \lambda_1 - \lambda_2 + 8 = 0$$

$$\Rightarrow 6\lambda_1 - 20\lambda_2 = 0$$

$$\Rightarrow \lambda_1 = \frac{10}{3} \lambda_2 \quad \text{--- ③}$$

Also,

$$7(\lambda_1 - 7\lambda_2 + 4) - 6(-2\lambda_1 + 6\lambda_2 + 6) + 1(\lambda_1 - \lambda_2 + 8) = 0$$

$$\Rightarrow 7\lambda_1 - 49\lambda_2 + 28 + 12\lambda_1 - 36\lambda_2 - 36 + \lambda_1 - \lambda_2 + 8 = 0$$

$$\Rightarrow 20\lambda_1 - 86\lambda_2 = 0$$

$$\Rightarrow \lambda_1 = \frac{43}{10} \lambda_2 \quad \text{--- ④}$$

Solving ③ & ④ we get $\lambda_1 = 0$ & $\lambda_2 = 0$

$\therefore P(3, 5, 7)$ & $Q(-1, -1, -1)$ are the two points where the S.D line intersect the lines

$\therefore$ Magnitude/length of the S.D line

$$= \sqrt{(3+1)^2 + (5+1)^2 + (7+1)^2}$$

$$= \sqrt{116}$$

$$= 2\sqrt{29}$$

Ans)

Equⁿ of the S.D line is given by

$$\frac{x-3}{3+1} = \frac{y-5}{5+1} = \frac{z-7}{7+1}$$

$$\Rightarrow \frac{x-3}{4} = \frac{y-5}{6} = \frac{z-7}{8}$$

$$\Rightarrow \frac{x-3}{2} = \frac{y-5}{3} = \frac{z-7}{4}$$

Problem:

Find the S.D between lines $\frac{x+3}{2} = \frac{y+5}{3} = \frac{z+7}{-3}$ and $\frac{x+1}{4} = \frac{y+1}{5} = \frac{z+1}{-1}$. Comment on whether the given SLs are coplanar or not.

Solⁿ:

Given SLs are:

$$\frac{x+3}{2} = \frac{y+5}{3} = \frac{z+7}{-3} \quad \text{--- (1)}$$

$$\& \frac{x+1}{4} = \frac{y+1}{5} = \frac{z+1}{-1} \quad \text{--- (2)}$$

Let (a, b, c) be the dir^s of the S.D line. Since the S.D line is perpendicular to both the lines (1) & (2), so we have

$$2a + 3b - 3c = 0 \quad \text{--- (3)}$$

$$\& 4a + 5b - c = 0 \quad \text{--- (4)}$$

Solving (3) & (4) using the cross-multiplication technique, we get

$$\frac{a}{-2+15} = \frac{b}{-12+2} = \frac{c}{10-12}$$

$$\Rightarrow \frac{a}{12} = \frac{b}{-10} = \frac{c}{-2}$$

$$\Rightarrow \frac{a}{6} = \frac{b}{-5} = \frac{c}{-1} = k \text{ (say)}$$

$$\therefore a = 6K, b = -5K, c = -K$$

let (l, m, n) be the d.c of the S.D line, then

$$l = \frac{a}{\sqrt{a^2 + b^2 + c^2}} = \frac{6k}{\sqrt{(6k)^2 + (-5k)^2 + (-k)^2}} = \frac{6k}{\sqrt{36 + 25 + 1}}$$

$$\therefore l = \frac{6}{\sqrt{62}}$$

Similarly,

$$m = \frac{-5}{\sqrt{62}}, \quad n = \frac{-1}{\sqrt{62}}$$

Any arbitrary point on line $\textcircled{D}$ is $P(-3, -5, -7)$

Any n n n n n ② is $Q(-1, -1, -1)$

$\therefore$ Length of the S.D line = Projection of the line joining the two points P & Q on the S.D. line

$$= |l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)|$$

$$= \left| \frac{6}{\sqrt{62}}(-1+3) - \frac{5}{\sqrt{62}}(-1+5) - \frac{1}{\sqrt{62}}(-1+7) \right|$$

$$= \frac{1}{\sqrt{62}} \left| 12 - 20 - 6 \right|$$

$$= \frac{14}{\sqrt{62}}$$

(Ans)

Since the S.D. between the lines ① & ② is not zero and the lines are not parallel, so the lines ① & ② are not coplanar.

Problem:

Find the magnitude and the eqn of the S.D line between the lines whose equations are

$$\frac{x-1}{4} = \frac{y-2}{3} = \frac{z-36}{-6} \text{ and } x+y=0, z=4$$

Soln:

Given the eqn's of SLs are

$$\frac{x-1}{4} = \frac{y-2}{3} = \frac{z-36}{-6} \text{ ————— ①}$$

$$\& \quad x+y=0, z=4 \quad \text{--- (2)}$$

Let us rewrite eqn (2) as follows

$$\left. \begin{aligned} x+y+0(z-4) &= 0 \\ 0 \cdot x+0 \cdot y+1 \cdot (z-4) &= 0 \end{aligned} \right\}$$

Hence, the symmetric form of the eqn (2) is now given by

$$\frac{x}{1} = \frac{y}{-1} = \frac{z-4}{0} \quad \text{--- (3)}$$

Let (a, b, c) be the dir^s of the S.D line

Since the S.D line is perpendicular to both the lines (1) & (3), so we have

$$4a+3b-6c=0 \quad \text{--- (4)}$$

$$\& \quad a-b=0 \quad \text{--- (5)}$$

From (4) & (5) using cross-multiplication we get

$$\frac{a}{0-6} = \frac{b}{-6} = \frac{c}{-7}$$

$$\Rightarrow \frac{a}{6} = \frac{b}{6} = \frac{c}{7} = k \text{ (say)}$$

$$\therefore a=6k, b=6k, c=7k$$

Let (l, m, n) be the d.c.s of the S.D. line, then

$$\begin{aligned} l &= \frac{a}{\sqrt{a^2+b^2+c^2}} = \frac{6k}{\sqrt{(6k)^2+(6k)^2+(7k)^2}} \\ &= \frac{6k}{\sqrt{36k^2+36k^2+49k^2}} \\ &= \frac{6}{11} \end{aligned}$$

Similarly, $m = \frac{6}{11}$ & $n = \frac{7}{11}$

Any arbitrary point on line ① is $P(1, 2, 36)$

" " " " " " ② " $Q(0, 0, 4)$

$\therefore$ Length of the S.D. line = Projection of the line joining the two points P & Q on S.D. line

$$\begin{aligned} &= \left| l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1) \right| \\ &= \left| \frac{6}{11}(1-0) + \frac{6}{11}(2-0) + \frac{7}{11}(36-4) \right| \\ &= \left| \frac{6+12+224}{11} \right| = 22 \text{ Units} \end{aligned}$$

Ans

Now, the eqn of the S.D line is given by

$$\begin{vmatrix} x-1 & y-2 & z-36 \\ 4 & 3 & -6 \\ 6 & 6 & 7 \end{vmatrix} = 0 = \begin{vmatrix} x & y & z-4 \\ 1 & -1 & 0 \\ 6 & 6 & 7 \end{vmatrix}$$

$$\Rightarrow 57(x-1) - 64(y-2) + 6(z-36) = 0$$

$$\Rightarrow 57x - 64y + 6z - 145 = 0$$

&

$$-7x - 7y + 12(z-4) = 0$$

$$\Rightarrow 7x + 7y - 12z + 48 = 0$$

$$\therefore 57x - 64y + 6z - 145 = 0 = 7x + 7y - 12z + 48$$

which is the required eqn's of the S.D line.

Problem:

Find the length and eqn of the S.D line between the two given lines $x+y+2z+5=0 = x+2y+2z+4$ and $x+2y+z+2=0 = x+y+3z+5$.

Soln:

Given,

$$x+y+2z+5=0 = x+2y+2z+4 \quad \text{--- (1)}$$

$$x+2y+z+2=0 = x+y+3z+5 \quad \text{--- (2)}$$

Let the eqn of the plane through the lines (1) is given by

$$x+y+2z+5 + k_1(x+2y+2z+4) = 0$$

$$\Rightarrow (1+k_1)x + (1+2k_1)y + (2+2k_1)z + 4k_1+5 = 0 \quad \text{--- (3)}$$

Similarly, the eqn of the plane through the lines (2) is given by

$$x+2y+z+2 + k_2(x+y+3z+5) = 0$$

$$\Rightarrow (1+k_2)x + (2+k_2)y + (1+3k_2)z + 5k_2+2 = 0 \quad \text{--- (4)}$$

If the planes (3) & (4) are parallel, then

$$\frac{1+k_1}{1+k_2} = \frac{1+2k_1}{2+k_2} = \frac{2+2k_1}{1+3k_2} = \lambda \text{ (say)}$$

$$\therefore 1+k_1 = \lambda(1+k_2) \Rightarrow 1+k_1 - \lambda - \lambda k_2 = 0 \text{ --- (5)}$$

$$1+2k_1 = \lambda(2+k_2) \Rightarrow 1+2k_1 - 2\lambda - \lambda k_2 = 0 \text{ --- (6)}$$

$$2+2k_1 = \lambda(1+3k_2) \Rightarrow 2+2k_1 - \lambda - 3\lambda k_2 = 0 \text{ --- (7)}$$

$$(6) - 2 \times (5)$$

$$-1 + \lambda k_2 = 0 \text{ --- (8)}$$

$$(7) - 2 \times (5)$$

$$\lambda - k_2 \lambda = 0$$

$$\Rightarrow k_2 \lambda = \lambda$$

$$\Rightarrow \boxed{k_2 = 1} \quad [\because \lambda \neq 0]$$

$$(8) \Rightarrow$$

$$\boxed{\lambda = 1}$$

Also,

$$\boxed{k_1 = 1}$$

Substituting the values of k_1, k_2 & λ into (3) & (4) we get respectively

$$2x + 3y + 4z + 9 = 0 \text{ ——— (9)}$$

$$\& \quad 2x + 3y + 4z + 7 = 0 \text{ ——— (10)}$$

The eqⁿs (9) and (10), i.e., the eqⁿs of the planes respectively pass through the lines represented by (1) & (2).

Since the planes (9) & (10) are parallel to each other so the distance between them itself is the S.D. (S.D)

$$\therefore \text{Length of the S.D line} = \left| \frac{d_1 - d_2}{\sqrt{a^2 + b^2 + c^2}} \right|$$

$$= \left| \frac{9 - 7}{\sqrt{2^2 + 3^2 + 4^2}} \right|$$

$$= \frac{2}{\sqrt{4 + 9 + 16}}$$

$$= \frac{2}{\sqrt{29}}$$

Ans

If (3) is perpendicular to either (9) or (10), then

$$2(1+k_1) + 3(1+2k_1) + 4(2+2k_1) = 0$$

$$\Rightarrow 2 + 2k_1 + 3 + 6k_1 + 8 + 8k_1 = 0$$

$$\Rightarrow 16k_1 + 13 = 0$$

$$\Rightarrow k_1 = -\frac{13}{16}$$

Substituting the value of k_1 into (3) we obtain

$$3x - 10y + 6z + 28 = 0$$

Again if (4) is perpendicular to either (9) or (10) then we have

$$2(1+k_2) + 3(2+k_2) + 4(1+3k_2) = 0$$

$$\Rightarrow 2 + 2k_2 + 6 + 3k_2 + 4 + 12k_2 = 0$$

$$\Rightarrow 17k_2 = -12$$

$$\Rightarrow k_2 = -\frac{12}{17}$$

Substituting the value of k_2 into (4) we obtain

$$5x + 22y - 19z - 26 = 0$$

Hence, the eqⁿs of the S.D line is given by

$$3x - 10y + 6z + 28 = 0 = 5x + 22y - 19z - 26$$

Ans

Problem:

Show that the length of the S.D between the lines $\frac{x+3}{-2} = \frac{y-6}{3} = \frac{z-3}{2}$ and $\frac{x}{2} = \frac{y-6}{-2} = \frac{z}{-1}$ is 3 and that the line of S.D cuts z-axis.

Solⁿ:

Given the eqⁿs of the SLs are

$$\frac{x+3}{-2} = \frac{y-6}{3} = \frac{z-3}{2} = k_1 \text{ (say)} \quad \text{--- (1)}$$

$$\frac{x}{2} = \frac{y-6}{-2} = \frac{z}{-1} = k_2 \text{ (say)} \quad \text{--- (2)}$$

Any arbitrary point on the line (1) is given by

$$P(-2k_1, -3, 3k_1+6, 2k_1+3)$$

Any arbitrary point on the line (2) is given by

$$Q(2k_2, -2k_2+6, -k_2)$$

Let (a, b, c) be the crs of the S.D line, then

$$\begin{aligned} (a, b, c) &\equiv (2k_2 + 2k_1 + 3, -2k_2 + 6 - 3k_1, -k_2 - 2k_1 - 3) \\ &\equiv (2k_2 + 2k_1 + 3, -2k_2 - 3k_1, -k_2 - 2k_1 - 3) \end{aligned}$$

Since, the S.D line is perpendicular to both the lines ① & ②, then

$$-2(2k_2 + 2k_1 + 3) + 3(-2k_2 - 3k_1) + 2(-k_2 - 2k_1 - 3) = 0$$

$$\Rightarrow -4k_2 - 4k_1 - 6 - 6k_2 - 9k_1 - 2k_2 - 4k_1 - 6 = 0$$

$$\Rightarrow -17k_1 - 12k_2 - 12 = 0$$

$$\Rightarrow 17k_1 + 12k_2 + 12 = 0 \quad \text{--- (3)}$$

Similarly,

$$2(2k_2 + 2k_1 + 3) - 2(-2k_2 - 3k_1) - 1(-k_2 - 2k_1 - 3) = 0$$

$$\Rightarrow 4k_1 + 3k_2 + 3 = 0 \quad \text{--- (4)}$$

Solving ③ & ④ we obtain

$$k_1 = 0 \text{ \& \> } k_2 = -1$$

Therefore, $P(-3, 6, 3)$ & $Q(-2, 8, 1)$ are the points where the S.D line meet at two respective planes.

$\therefore$ S.D = Distance between the two points
 $P(-3, 6, 3)$ & $Q(-2, 8, 1)$

$$\Rightarrow S.D = \sqrt{(-2+3)^2 + (8-6)^2 + (1-3)^2}$$

$$= \sqrt{1+4+4}$$

$$= 3$$

[Showed]

Now, eqnⁿ of the S.D line be

$$\frac{x+3}{-1} = \frac{y-6}{-2} = \frac{z-3}{2} = k \text{ (say)}$$

$\therefore$ Any arbitrary point on the S.D line be

$$M(-k-3, -2k+6, 2k+3)$$

Again let the eqnⁿ of the z-axis be

$$\frac{x}{0} = \frac{y}{0} = \frac{z}{1} = \lambda \text{ (say)}$$

$\therefore$ Any arbitrary point on z-axis be

$$N(0,0,\lambda)$$

If S.D line & z-axis intersect with each other then

$$-k-3=0 \Rightarrow k=-3$$

$$-2k+6=0 \Rightarrow k=-3$$

$$2k+3=\lambda \Rightarrow \lambda=-3$$

Hence, the coordinates of the point on the z-axis be $(0, 0, -z)$

Thus, S.D line intersects z-axis.

[Shown]

Problem:

Show that the eqn of the plane passing through the points $(2, -1, 0)$ and $(3, -4, 5)$ and parallel to the straight line $2x = 3y = 4z$ is given by

$$29x - 27y - 22z - 85 = 0.$$

Soln:

Let the eqn of the plane passing through the point $(2, -1, 0)$ is given by

$$a(x-2) + b(y+1) + c(z-0) = 0 \quad \text{--- (1)}$$

Also, the eqn of the straight line is

$$2x = 3y = 4z$$

$$\Rightarrow \frac{x}{6} = \frac{y}{4} = \frac{z}{3} \quad \text{--- (2)}$$

The plane (1) is parallel to the SL (2), therefore, the normal to the plane (1) is perpendicular to (2)

$$\therefore 6a + 4b + 3c = 0 \text{ ————— } \textcircled{3}$$

Again, plane ① passes through the point $(3, -4, 5)$, therefore

$$a(3-2) + b(-4+1) + c(5-0) = 0$$

$$\Rightarrow a - 3b + 5c = 0 \text{ ————— } \textcircled{4}$$

Eliminating a, b, c from eqn's ①, ③, ④

gives

$$\begin{vmatrix} x-2 & y+1 & z \\ 6 & 4 & 3 \\ 1 & -3 & 5 \end{vmatrix} = 0$$

$$\Rightarrow (x-2)(20+9) + (y+1)(2-30) + z(-18-4) = 0$$

$$\Rightarrow 29(x-2) - 27(y+1) - 22z = 0$$

$$\Rightarrow 29x - 27y - 22z - 58 - 27 = 0$$

$$\Rightarrow 29x - 27y - 22z - 85 = 0$$

[Showned]

Problem:

Prove that the equation to the plane containing the line $\frac{y}{b} + \frac{z}{c} = 1, x=0$ and parallel to the line

$\frac{x}{a} - \frac{z}{c} = 1, y=0$ is $\frac{x}{a} - \frac{y}{b} - \frac{z}{c} + 1 = 0$ and if $2d$ is the shortest distance (S.D) then also prove that

$$\frac{1}{d^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}.$$

Solⁿ:

Let the equation of the plane containing the SL

$\frac{y}{b} + \frac{z}{c} = 1, x=0$ is the given by

$$\left(\frac{y}{b} + \frac{z}{c} - 1\right) + \lambda x = 0 \quad \text{--- (1a)}$$

$$\Rightarrow \lambda x + \frac{1}{b} y + \frac{1}{c} z - 1 = 0 \quad \text{--- (1b)}$$

Given, $\frac{y}{b} + \frac{z}{c} = 1, x=0$ --- (1)

& $\frac{x}{a} - \frac{z}{c} = 1, y=0$ --- (2)

The symmetric form of the eqnⁿ (2) is given

by $\frac{x}{a} = \frac{y}{0} = \frac{z+c}{c}$ --- (3)

Since (3) is parallel to (1b), then we have

$$ak + 0 \cdot \frac{1}{b} + c \cdot \frac{1}{c} = 0$$

$$\Rightarrow ak + 1 = 0$$

$$\Rightarrow k = -\frac{1}{a}$$

Substituting the value of k into (1b) to have

$$-\frac{1}{a}x + \frac{1}{b}y + \frac{1}{c}z - 1 = 0$$

$$\Rightarrow \frac{1}{a}x - \frac{1}{b}y - \frac{1}{c}z = -1$$

$$\Rightarrow \frac{x}{a} - \frac{y}{b} - \frac{z}{c} + 1 = 0 \quad \text{--- (4)}$$

[Proved]

From (3) we can write,

$$\frac{x}{a} = \frac{y}{0} = \frac{z+c}{c} = \lambda \text{ (say)}$$

$$\therefore x = a\lambda, \quad y = 0, \quad z = c\lambda - c$$

For $\lambda = 1$, $(a, 0, 0)$ be any point on the line (3)

Let $2d$ be the S.D from any arbitrary point

$(a, 0, 0)$ to the plane (4), then we have

$$2d = \left| \frac{\frac{a}{a} + 0 + 0 + 1}{\sqrt{\left(\frac{1}{a}\right)^2 + \left(-\frac{1}{b}\right)^2 + \left(\frac{1}{c}\right)^2}} \right|$$

$$\Rightarrow 2d = \frac{2}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

$$\Rightarrow 4d^2 = \frac{4}{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}$$

$$\Rightarrow d^2 = \frac{1}{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}$$

$$\Rightarrow \frac{1}{d^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$$

[Proved]